

PRASAD JAWALE

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EDUCATION

- University of Wisconsin - Madison:** *Master of Science in Data Science* Sept 2024 – May 2026
Coursework: Optimization, Big Data Systems, Machine Learning, Artificial Intelligence
- University of Mumbai:** *Bachelor of Technology in Artificial Intelligence and Data Science* Oct 2020 – May 2024
Coursework: Database Management Systems, Algorithms, Operating Systems, Computer Networks

EXPERIENCE

- Data Analyst Intern, Wisconsin School of Business** Oct 2024 – Mar 2025
- Processed 100K+ hotel reviews into structured data by building a Python and SQL ETL pipeline with efficient text preprocessing and extraction.
 - Improved multilingual spam classification to an F1-score of 0.82 by implementing a model using text embeddings and weak supervision.
 - Identified long-term customer sentiment trends over a decade by applying LDASeq and BERTopic for topic modeling, supporting insights into evolving feedback patterns.
- Software Engineer Intern (Android), StocCGuru** Jun 2021 – Aug 2021
- Optimized UI performance by implementing RecyclerView data binding and OkHttp REST APIs, converting Figma designs into responsive, dynamic Android components.
 - Accelerated deployment by developing Flask API unit and integration tests using Postman, ensuring 100% backend reliability during Agile development sprints.

PROJECTS

- Credit Risk Inference Service - [Github Link](#)**
- Delivered real-time credit risk predictions with 450ms p99 latency by engineering a FastAPI-based inference service with request validation using Pydantic.
 - Reduced manual deployment and integration effort by 60% by designing a CI/CD pipeline using GitHub Actions and Docker Compose to automate testing and ECR-to-EC2 deployments.
 - Improved model lifecycle efficiency and retrieval time by integrating MLflow with PostgreSQL (Amazon Aurora) and S3 for experiment tracking, model registry, and artifact storage.
- Semantic Document Retriever - [Github Link](#)**
- Enabled search and retrieval over 10K+ documents with 92% top-5 retrieval accuracy by building a full-stack system with a React front-end and FastAPI back-end that supports structured querying.
 - Reduced latency and improved relevance using an Elasticsearch hybrid search pipeline with Reciprocal Rank Fusion-based reranking.
 - Achieved 87% accuracy on benchmark QA tasks by developing a RAG-based question-answering system using Gemini LLM and LangChain over the document corpus.
- Collaborative Filtering Music Recommender - [Github Link](#)**
- Scaled recommendation training over 6M+ tracks and 1M playlists by building a distributed PySpark ETL pipeline to process and transform Spotify playlist JSON data into structured datasets.
 - Reduced memory usage and enabled efficient similarity computations by converting playlist-track data into sparse matrix representations for KNN-based collaborative filtering.
 - Achieved over 80% top-10 recommendation hit rate by implementing a similarity-based recommender using nearest neighbor search on transformed playlist embeddings.
- Pharmaceutical Logistics Optimization - [Github Link](#)**
- Improved shipment mode selection efficiency and optimized cost vs delivery-time trade-offs by building an end-to-end decision system using machine learning models trained on logistics data.
 - Automated model retraining and ETL workflows for 86K+ records by orchestrating pipelines using Apache Airflow with scheduled DAGs and data validation steps.
 - Achieved a Random Forest model with 0.82 AUC by performing hyperparameter optimization using Optuna and iterative experimentation on engineered features.

SKILLS

Languages: Python, SQL, C/C++, Bash/Linux
Backend & APIs: FastAPI, Flask, Postman
Machine Learning: PyTorch, Scikit-Learn, HuggingFace, Random Forests, Optuna, BERTopic
Infrastructure & Data: Apache Airflow, Docker, AWS (EC2, S3, RDS), PostgreSQL, PySpark MongoDB, GitHub Actions